

Revolutionizing Recruitment: Enhanced Machine Learning Models for Bias Mitigation and Efficiency

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Abstract— *This study investigates the enhanced application of machine learning models to optimize recruitment processes by predicting hiring decisions based on comprehensive candidate profiles. Utilizing an expanded dataset of 1,500 candidates, the research incorporates diverse features such as age, gender, education level, work experience, skill scores, social media activity, and psychometric assessments. Several machine learning algorithms, including Random Forest, Support Vector Machine, Logistic Regression, and advanced ensemble methods (CatBoost, XGBoost), were evaluated to determine the best-performing model and identify the most influential features. Results show that CatBoost outperformed other models, achieving an accuracy of 95%, followed by Random Forest and XGBoost. Analysis of feature importance revealed that Recruitment Strategy, Education Level, and Personality Score were the top three factors influencing hiring decisions.*

The study further enhances the model by implementing deep learning techniques and fairness-aware algorithms to mitigate biases. Bias detection and correction techniques, along with fairness metrics such as demographic parity and equal opportunity, were employed to ensure equitable hiring practices. Explainable AI methods, including SHAP and LIME, were incorporated to improve model interpretability for HR professionals. Additionally, the study explores the integration of ethical guidelines and data privacy measures to ensure transparency and accountability in AI-driven recruitment processes. These enhancements demonstrate the potential of machine learning to streamline recruitment, reduce biases, and improve hiring outcomes, promoting more efficient and equitable recruitment processes.

Keywords— *Machine Learning, Deep Learning, Explainable AI, Predictive Hiring, Recruitment, SHAP, LIME, Hyperparameter Tuning, Classification, Fairness-aware algorithm, Bias Mitigation & Modelling*

I. INTRODUCTION

Recruitment is a critical function for organizations, directly influencing workforce quality, diversity, and alignment with strategic goals. Traditionally, hiring processes have relied heavily on manual screening, interviews, and subjective judgment, which can introduce inefficiencies and biases [1]. These traditional methods, while familiar and intuitive, often fall short in evaluating candidates holistically, especially when faced with high application volumes and time constraints [2]. In response, organizations have increasingly turned to machine learning (ML) to automate and optimize various stages of the hiring process, including resume screening, candidate ranking, and decision support [3].

While ML models offer significant potential to enhance hiring accuracy and efficiency, their implementation in recruitment raises complex ethical and practical challenges. Algorithms trained on historical data may inadvertently learn and perpetuate societal and institutional biases—such as those related to gender, age, ethnicity, or educational background—leading to discriminatory hiring outcomes [4][5]. Moreover, many high-performing models function as opaque "black boxes," making their decisions difficult to interpret for HR professionals and candidates alike [6]. These issues have prompted a shift in academic and industry discourse from mere prediction accuracy to broader considerations of fairness, accountability, transparency, and ethics in algorithmic decision-making [7].

This study addresses these emerging concerns by proposing an enhanced ML framework for recruitment that not only predicts hiring outcomes but also integrates fairness-aware learning, explainable AI techniques, and ethical safeguards. By combining traditional and ensemble models with deep learning architectures, the study evaluates predictive performance, fairness metrics, and interpretability. The inclusion of SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) provides transparency into the model's decision-making process, while fairness-aware algorithms such as Exponentiated Gradient ensure that bias mitigation is actively pursued.

A. Problem Statement

Although machine learning has demonstrated considerable promise in improving recruitment accuracy, existing models often neglect issues related to fairness, interpretability, and accountability. Training data frequently reflect existing societal inequalities, which ML models can inherit and amplify [8]. For instance, a dataset overrepresenting one gender or academic background can bias the model toward those groups, perpetuating historical disparities in hiring [9]. Furthermore, reliance on black-box models limits transparency and makes it difficult for HR professionals to trust or justify automated decisions, particularly in contexts where compliance with equal opportunity legislation is crucial [10]. While some prior studies have compared the predictive performance of ensemble methods like XGBoost and CatBoost, few have rigorously integrated fairness audits, explainability tools, and ethical AI design into a cohesive recruitment pipeline.

B. Objectives

The primary objective of this research is to develop and evaluate a comprehensive machine learning framework for recruitment that balances predictive performance with fairness, transparency, and ethical responsibility. Specifically, the study seeks to:

- 1) *Model Comparison*: Evaluate the predictive performance of traditional (e.g., Logistic Regression), ensemble (e.g., Random Forest, CatBoost, XGBoost), and deep learning models.
- 2) *Fairness Metric Analysis*: Assess bias across sensitive attributes (e.g., gender) using fairness metrics such as demographic parity difference and equalized odds difference.
- 3) *Bias Mitigation Techniques*: Implement and evaluate fairness-aware algorithms (e.g., Exponential Gradient) to mitigate bias in model outcomes.
- 4) *Explainable AI*: Apply interpretability methods such as SHAP and LIME to explain model decisions and promote transparency.

C. Significance of this Study

This research extends existing literature by bridging the gap between algorithmic performance and responsible AI in recruitment. While prior studies have primarily focused on enhancing model accuracy, this work integrates fairness-aware algorithms and explainability tools to ensure equitable and interpretable outcomes [11]. The proposed framework aligns with recent calls from scholars and policymakers to ensure that AI systems employed in high-stakes decision making adhere to principles of fairness, accountability, and transparency [12][13]. In doing so, it offers actionable insights for both academic researchers and HR practitioners aiming to implement AI in a socially responsible manner.

D. Research Question

This central research question guiding this study is: **How can machine learning models be enhanced to optimize recruitment decisions while ensuring fairness, transparency, and efficiency?** To address this question, the study analyses features such as education, skill scores, and work experience, with the goal of developing models that facilitate automated, equitable, and effective hiring processes:

In addition to the primary research question, this study also investigates the following sub-questions:

- 1) Which candidate features are most predictive of hiring outcome?
- 2) How do traditional, ensemble, and deep learning models compare in terms of predictive accuracy and fairness?
- 3) Can fairness-aware algorithms effectively mitigate bias in hiring predictions?
- 4) How can SHAP and LIME enhance the interpretability of model decisions for HR professionals.

E. Literature Review

The application of machine learning in recruitment has garnered increasing attention due to its potential to automate candidate screening, reduce time-to-hire, and improve hiring outcomes. Early implementations primarily focused on resume parsing and candidate ranking, often utilizing supervised learning methods such as logistic regression and decision trees [2]. As ML techniques matured, ensemble methods like Random Forest, Gradient Boosting, and CatBoost gained prominence for their ability to model complex, non-linear relationships and enhance predictive accuracy [14].

However, a growing body of research has raised concerns about fairness and discrimination in algorithmic hiring. Barocas et al. [5] argue that predictive models trained on historical hiring data can inadvertently learn and perpetuate past biases, especially when those biases are embedded in features such as education, gender, or employment history. Cowgill et al. [9] found that although algorithms may outperform humans in predictive

accuracy, they often replicate discriminatory patterns unless fairness is explicitly addressed. In response, researchers have incorporated fairness metrics and bias mitigation techniques into ML workflows. Approaches such as reweighing, adversarial debiasing, and constraint-based optimization (e.g., demographic parity, equalized odds) are increasingly adopted to ensure equitable outcomes [15].

Peña et al. [16] proposed a multimodal framework leveraging structured, image, and text data to assess candidates. Although the model demonstrated technical sophistication, it lacked real-time interpretability—an essential factor for HR professionals seeking transparency in AI-driven decisions. The authors recommended integrating tools such as SHAP and LIME to improve model explainability. Similarly, Tasheva and Karpovich [12] emphasized the importance of balancing automation with human oversight and highlighted critical concerns related to algorithmic opacity, ethical compliance, and candidate privacy.

In a comparative study of recruitment models, Al-Quhfa et al. [14] reported that ensemble methods and neural networks yielded strong predictive performance but failed to incorporate fairness or explainability mechanisms. They advocated for the development of end-to-end frameworks that combine fairness-aware training with interpretable outputs.

Taken together, these studies underscore the need for holistic machine learning frameworks that go beyond accuracy to also address fairness, transparency, and accountability. This research builds on prior work by incorporating SHAP, LIME, and fairness-aware algorithms into a unified recruitment prediction pipeline—thereby addressing a critical gap in the literature.

II. METHODOLOGIES

A. Dataset Summary

The dataset utilized in this study comprises detailed profiles of 1,500 candidates, each annotated with 11 structured features and a binary target variable indicating hiring decision (1 for hired, 0 for not hired). The features were selected based on common recruitment evaluation criteria and reflect a blend of demographic, academic, professional, and behavioural dimensions relevant to the hiring process. These include age, gender, educational background, years of experience, and skill scores. Table I outlines the features and their respective data types.

TABLE I
SUMMARY OF THE FEATURES IN THE DATASETS

Feature	Description	Data Type
Age	Age of the Candidate	Numerical
Gender	0: Male, 1: Female	Categorical
Education Level	1: Bachelor's (Type 1), 2: Bachelor's (Type 2), 3: Master's, 4: PhD	Categorical
Experience Years	Number of years of professional experience	Numerical
Interview Scores	Score achieved by the candidate in the interview process	Numerical
Skill Score	Assessment score of candidate's technical skills	Numerical
Personality Score	Evaluation score of the candidate's personality traits	Numerical
Recruitment Strategy	1: Aggressive, 2: Moderate, 3: Conservative	Categorical
Hiring Decision	0: Not Hired, 1: Hired	Categorical

B. Preprocessing Steps

1) *Encoding Categorical Variables*: In the preprocessing phase, some variables with discrete, limited values were recast as categorical variables to enhance model interpretability. For example, Gender, Education Level, and Recruitment Strategy were originally represented numerically (e.g., 0 for male, 1 for female) and were redefined as categorical labels for clearer classification during model training. This conversion helps improve the model's ability to distinguish between categories effectively and allows for better handling by certain machine learning algorithms.

2) *Handling Missing Values & Outliers*: Initial data validation revealed no null values or duplicated records. However, boxplot analysis identified a few high-end outliers in Interview, Skill, and Personality scores. These were retained intentionally, as they represented exceptional candidates rather than errors. Their inclusion added realism and preserved the natural variability in candidate profiles.

3) *Feature Scaling*: Feature scaling was applied selectively. For distance-based or gradient-sensitive models (SVM, DNN), z-score normalization ensured standardized inputs. Tree-based models were trained on raw feature distributions, leveraging their scale invariance and ability to handle skewed data.

4) *Cross Validation & Data Splitting*: To maximize generalizability, the study employed 30-fold stratified cross-validation, ensuring each fold preserved the original class distribution. Given the dataset of 1,500 candidates, using 30 folds ensured that each validation set contained approximately 50 data points, maintaining

a balance between training and validation splits without losing too much training data. A larger K value helps reduce variance by creating smaller validation sets, leading to a more reliable estimate of model performance across different data subsets. By choosing K=30, we minimized the impact of outliers and enhanced the model's ability to generalize effectively. Hyperparameter tuning was also conducted via grid search within cross-validation loops, enhancing reproducibility.

5) *Exploratory Analysis:* Table II presents a summary of the key statistics for the candidates' demographic and performance metrics. The mean age of participants is 35.15 years, with a standard deviation (SD) of 9.25 years, indicating moderate variation in age among respondents. The median age is 35 years, closely aligned with the mean, suggesting a fairly symmetrical distribution. The mode is 45 years, indicating that this was the most frequently occurring age in the dataset. The minimum age recorded is 20 years, while the maximum is 50 years.

For work experience, the mean is 7.69 years, with an SD of 4.64 years, showing some variability in the dataset. The median experience is 8 years, indicating that candidates tend to have senior-level experience. The mode is 15 years, the most common value recorded. The candidates' years of experience ranges from 0 to 15 years.

The interview scores have a mean of 50.56, with a high SD of 28.62, reflecting considerable variability in candidate performance. The median score is 52, slightly above the mean, and the mode is 72, indicating that this score appeared most frequently. Scores range from 0 to 100, representing the full spectrum of possible outcomes. Similarly, the skill scores have a mean of 51.12 and an SD of 29.35, highlighting a broad distribution. The median score is 53, and the mode is 79, suggesting that this value was the most observed. The minimum and maximum skill scores are 0 and 100, respectively.

The personality scores have a mean of 49.38 and a standard deviation (SD) of 29.35, closely resembling the distribution pattern of skill scores. The median score is 49, and the mode is 43, indicating that 43 was the most frequently observed score. The range ranges from 0 to 100, demonstrating wide variability in personality assessments.

These statistics highlight variability across all variables, particularly in interview, skill, and personality scores, where standard deviations are relatively high. The presence of minimum values at zero suggests some participants scored poorly in these categories, whereas others achieved perfect scores, indicating diverse performance levels among respondents.

TABLE II
SUMMARY OF THE FEATURES IN THE DATASETS

Statistics	Age (years)	Experience Years	Interview Scores	Skill Score	Personality Score
Mean	35.15	7.69	50.56	51.12	49.38
SD	9.25	4.64	28.62	29.35	29.35
Median	35	8	52	53	49
Mode	45	15	72	79	43
Minimum	20	0	0	0	0
Maximum	50	15	100	100	100

6) *Correlation Heatmap:* Figure 1 presents the correlation heatmap for the dataset's features with the target variable, HiringDecision. According to Figure 1 and Table III, SkillScore, PersonalityScore and InterviewScore have the highest positive correlation with the target feature, HiringDecision, indicating that higher scores in these areas are associated with a higher likelihood of a positive hiring decision.

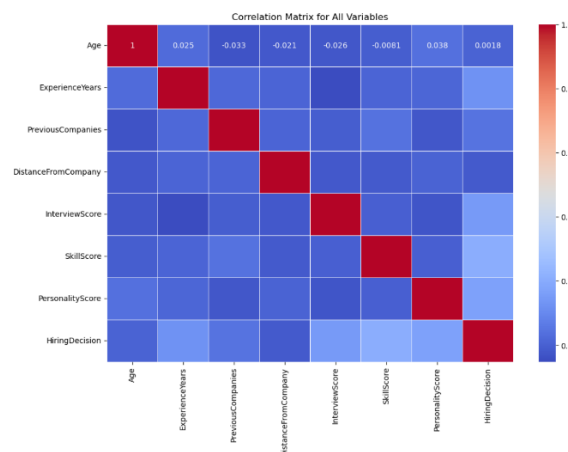


Fig. 1 Correlation Heatmap for All Numerical Variables

TABLE III
STRENGTH OF CORRELATION WITH THE TARGET VARIABLE

Features	Correlation Score
SkillScore	0.2037
PersonalityScore	0.1692
InterviewScore	0.1461
ExperienceYears	0.1225
PreviousCompanies	0.0441
Age	0.0185
DistanceFromCompany	-0.0168

7) *Gender Distribution in the Dataset*: Figure 2 shows a near-equal gender distribution in the dataset, with males comprising 50.8% of the data and females making up 49.2%. This balanced gender representation is important for ensuring fairness in predictive modeling and reducing gender biases in hiring predictions rather than errors. Their inclusion added realism and preserved the natural variability in candidate profiles.

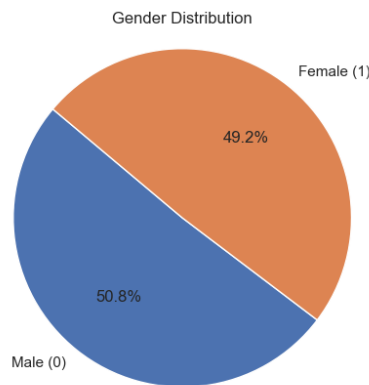


Fig. 1 Gender Distribution

8) *Education Level Distribution*: Figure 3 shows the number of candidates by education level. The majority, around 740 candidates, have a Bachelor's (Type 2) degree. Master's degrees are held by 317 candidates, while 307 candidates have a Bachelor's (Type 1) degree. The least common are PhD holders, with approximately 136 candidates, indicating fewer highly specialized individuals. scaling was applied selectively.

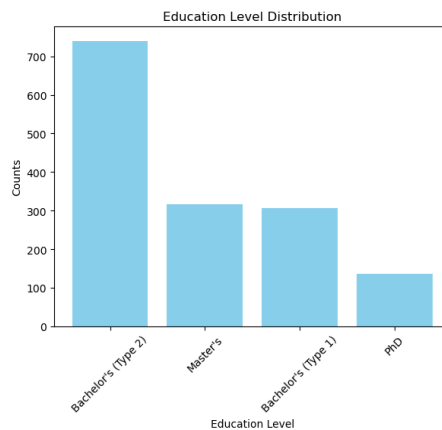


Fig. 3 Education Level Distribution

9) *Recruitment Strategy Distribution*: Figure 4 displays the distribution of different recruitment strategies used in the dataset. The "Moderate" strategy is the most prevalent, applied to approximately 770 candidates, indicating a balanced approach that weighs both candidate potential and organizational requirements. The "Aggressive" strategy, implemented for around 450 candidates, suggests a more proactive hiring approach, likely prioritizing rapid talent acquisition to meet urgent workforce demands. In contrast, the "Conservative" strategy, used for roughly 280 candidates, reflects a more selective hiring process, potentially favoring highly qualified or specialized candidates. This distribution highlights a clear preference for a balanced hiring approach over extreme selectivity, suggesting that organizations aim to optimize efficiency while maintaining a thorough

evaluation process. The varied use of strategies may also indicate differences in hiring needs across job roles, industries, or company cultures.



Fig. 4 Recruitment Strategy Distribution

C. Machine Learning Models and Fairness Techniques

To ensure a thorough evaluation of the factors influencing hiring decisions, this study employed a range of machine learning algorithms. These models were selected for their ability to handle complex, structured datasets and to mitigate overfitting, as follows:

1) *Logistic Regression (Baseline Model)*: Logistic Regression (LR) was used as a baseline model due to its simplicity and interpretability, and ability to output probabilities. Regularization was applied to prevent overfitting

2) *Random Forest (RF)*: Random Forest was chosen for its robustness in handling high-dimensional data and its ability to mitigate overfitting by averaging multiple decision trees. It is particularly useful in handling complex relationships between features and can provide feature importance rankings [17].

3) *Support Vector Machine (SVM)*: SVM was employed to test the model's ability to classify candidates based on a linear or non-linear decision boundary. SVM is well-suited for datasets where the decision boundary is not easily separable and can perform well with high dimensional data.

4) *Gradient Boosting Algorithms (XGBoost, LightGBM, CatBoost, AdaBoost)*: These ensemble methods were included due to their strong predictive power and efficiency in handling large datasets. Gradient boosting algorithms iteratively improve model predictions by correcting errors from previous iterations. XGBoost, LightGBM, and CatBoost were specifically chosen for their ability to handle imbalanced data and capture complex non-linear relationships [18][19].

5) *Deep Neural Network (DNN)*: A multi-layer perceptron with ReLU activations, dropout regularization, and a sigmoid output was trained on standardized features. Though less interpretable, it captured deeper nonlinear interactions among features.

6) *Fairness-Aware Algorithm*: Exponentiated Gradient Implemented using Fairlearn's reduction interface. The model was trained under a demographic parity constraint, and outcomes were compared with the base Logistic Regression model. Results demonstrated trade-offs between fairness improvement and slight reductions in accuracy, consistent with prior fairness research [15][16].

7) *Model Explainability*: SHAP and LIME - SHAP values were calculated for CatBoost and XGBoost to provide both global and instance-level feature attributions. SHAP summary plots ranked predictors such as Recruitment Strategy, Personality Score, and Education Level as top contributors. LIME offered localized explanations for individual predictions and was particularly useful for auditing edge cases and model behavior on minority groups.

III. RESULTS

A. Model Accuracy Scores

Figure 5 illustrates the overall accuracy scores for all machine learning models evaluated in this study. The CatBoost classifier demonstrated the highest accuracy at 95.4%, indicating its superior capacity to learn complex relationships and effectively handle categorical variables. Random Forest and XGBoost also performed competitively with accuracy scores of 94.4% and 94.2%, respectively. Gradient Boosting and LightGBM

achieved 93%, while Extra Trees, SVM, and Deep Neural Network (DNN) models maintained strong consistency with accuracies of 92%. AdaBoost reached 91%. Logistic Regression, the baseline model, while offering interpretability, lagged in predictive power, recording an accuracy of 87%. These results reinforce the benefit of using ensemble and deep learning models for recruitment prediction tasks, demonstrating the advantage of ensemble learning and deep architectures in modelling complex hiring decisions over more linear, traditional classifiers.

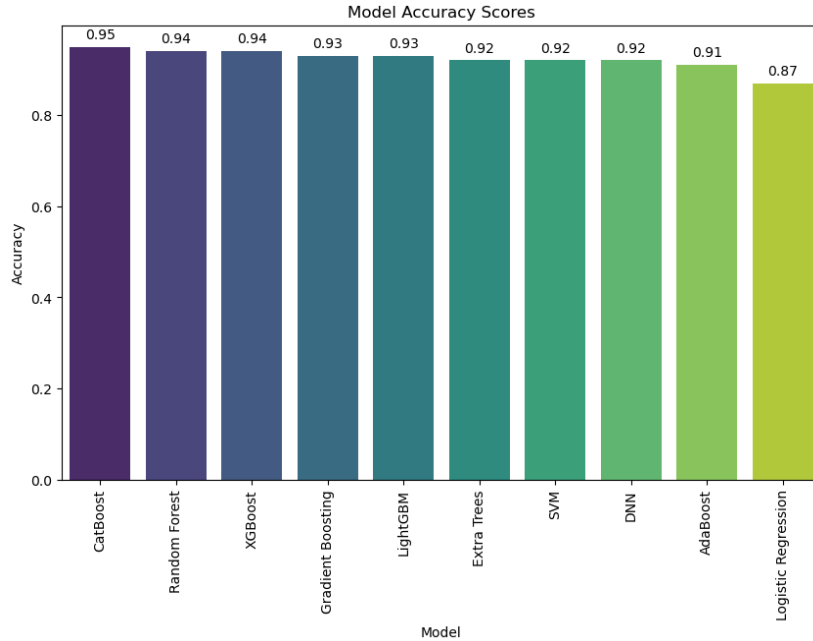


Fig. 5 Histogram of Model Accuracy Scores

B. Model Comparison and Evaluation

1) *CatBoost*: Table IV. shows the classification report for CatBoost. This model achieves impressive results with a high precision of 0.96 for class 0 (Not Hired) and 0.93 for class 1 (Hired). The recall for class 1 (0.91) and class 0 (0.97) suggests that CatBoost performs well in both identifying hires and non- hires, with an overall accuracy of 0.95.

TABLE IV.
CLASSIFICATION REPORT FOR CATBOOST

	Precision	Recall	F1 Score	Recall
Class 0	0.96	0.97	0.97	215
Class 1	0.93	0.91	0.92	85
Accuracy			0.95	300

2) *Random Forest (RF)*: Table V. shows the classification report for Random Forest. The model exhibits a high precision of 0.95 for class 0 and 0.94 for class 1, but the recall for class 1 is slightly lower (0.86), suggesting it may miss some hires. Despite this, it maintains an accuracy of 0.94, indicating strong overall performance.

TABLE V.
CLASSIFICATION REPORT FOR RANDOM FOREST

	Precision	Recall	F1 Score	Recall
Class 0	0.95	0.98	0.96	215
Class 1	0.94	0.86	0.90	85
Accuracy			0.94	300

3) *XGBoost*: Table VI. shows the classification report for XGBoost. This model exhibits strong precision and recall for class 0 (0.95 and 0.96, respectively). However, its performance for class 1 (hired) is slightly lower with precision of 0.90 and recall of 0.88, suggesting it may miss more positive instances. Nevertheless, its overall accuracy remains 0.94.

TABLE VIII.
CLASSIFICATION REPORT FOR XGBOOST

	Precision	Recall	F1 Score	Recall
Class 0	0.95	0.98	0.96	215
Class 1	0.94	0.86	0.90	85
Accuracy			0.94	300

C. ROC-AUC Curves

Figure 6 presents the Receiver Operating Characteristic (ROC) curves for all machine learning models, highlighting their ability to distinguish between hired and non-hired candidates. The Area Under the Curve (AUC) metric, with values closer to 1 indicating superior classification performance, serves as an indicator of overall discriminative power. CatBoost, Random Forest, AdaBoost, Extra Trees, and Gradient Boosting all exhibited AUC scores of 0.93, demonstrating highly effective separation of the two classes across various thresholds. These models maintain consistent predictive quality, managing false positives and false negatives effectively, which is crucial in high-stakes decisions like hiring. SVM, Logistic Regression, XGBoost, LightGBM, and DNN yielded slightly lower but still strong AUC scores of 0.92, suggesting that while competent, these models may show slightly higher variability in prediction certainty under different threshold settings. CatBoost, with the highest ROC-AUC score of 0.9292, demonstrates superior performance in distinguishing between hired and non-hired candidates.

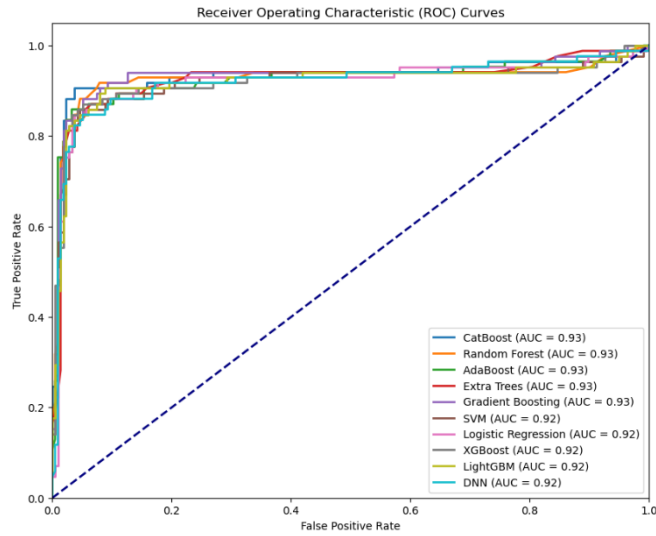


Fig. 6 ROC Curves for all Models

D. Hyperparameter Tuning

Hyperparameter tuning was performed using GridSearchCV and RandomSearchCV to optimize model performance. Table VII. indicates that CatBoost experienced a slight improvement, affirming the model's ability to perform well even with default settings. However, XGBoost and Random Forest showed marginal decreases in accuracy post-tuning, suggesting potential overfitting or plateauing performance. Logistic Regression remained relatively stable.

TABLE VII.
MODEL ACCURACY AFTER HYPERPARAMETER TUNING

Model	Before Tuning	After Tuning (GridSearchCV)	After Tuning (RandomSearch)
CatBoost	0.9533	0.9542	0.9542
XGBoost	0.9400	0.9292	0.9292
Random Forest	0.9400	0.9242	0.9225
Logistic Regression	0.8667	0.8657	0.8558

E. Fairness Evaluation and Bias Mitigation

To ensure ethical integrity in hiring decisions, this study employed a fairness-aware evaluation using gender as the sensitive attribute. The baseline logistic regression model exhibited fairness concerns, with a Demographic Parity Difference (DPD) of 0.0142 and an Equalized Odds Difference (EOD) of 0.0414, suggesting measurable disparities in predictions across gender groups. To address this, the model was retrained using the Exponentiated Gradient algorithm with a Demographic Parity constraint, resulting in notable

improvements: accuracy increased to 88.16% for male candidates and 93.24% for female candidates, while DPD and EOD dropped to approximately 0.01 and 0.03, respectively. These outcomes demonstrate that fairness-aware retraining can effectively mitigate bias while preserving predictive performance, thereby enhancing the ethical reliability of AI-driven hiring systems.

F. SHAP (SHapley Additive exPlanations)

SHAP analysis revealed that RecruitmentStrategy was by far the most influential feature in the model, with a SHAP impact score of 1.8872. This suggests that the strategic approach employed by the organization (e.g., aggressive vs. conservative recruitment) heavily influenced the model's decision-making process. EducationLevel and SkillScore followed with impact values of 0.8064 and 0.8038, respectively, indicating their strong contributions to positive hiring decisions. Other notable contributors included PersonalityScore (0.7793), ExperienceYears (0.6980), and InterviewScore (0.6159). These findings are aligned with common hiring criteria in real-world HR contexts, where both cognitive and interpersonal traits are weighted alongside academic and technical achievements.

Less influential features included PreviousCompanies (0.1902), DistanceFromCompany (0.1140), Gender (0.1004), and Age (0.0984), all of which showed relatively low SHAP values. The low importance assigned to sensitive attributes like Gender and Age is particularly encouraging from a fairness and compliance standpoint, suggesting that the model learned to base hiring decisions on meritocratic signals rather than demographic traits.

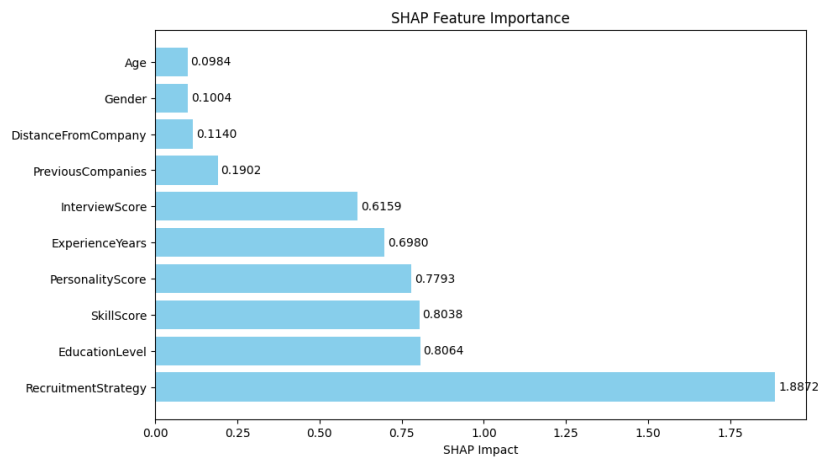


Fig. 7 SHAP Feature Importance

G. LIME (Local Interpretable Model-Agnostic Explanations)

According to Figure 8, SkillScore emerged as the most impactful local feature (importance: 0.1575), followed by PreviousCompanies (0.0655), Gender (0.0565), and DistanceFromCompany (0.0501). Interestingly, while RecruitmentStrategy was the dominant feature in SHAP, it ranked much lower in LIME (0.0039), suggesting that its influence is more structural or globally learned, rather than highly variable on a per-candidate basis.

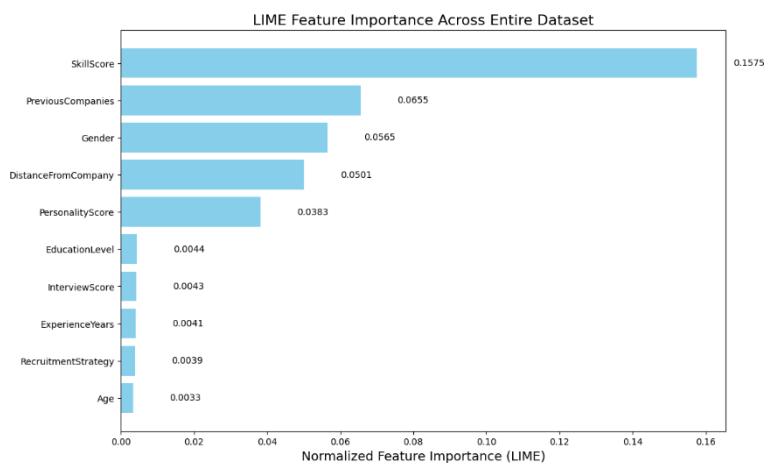


Fig. 8 LIME Feature Importance

IV. DISCUSSIONS

A. Model Accuracy and Performance

The consistent high accuracy scores obtained across multiple models affirm the feasibility of using machine learning to support hiring decisions. CatBoost emerged as the top-performing model, achieving an accuracy of 95.4%, followed closely by Random Forest and XGBoost. These values exceed the standard benchmarks for classification performance and suggest that ensemble-based models are well-suited for modeling complex, nonlinear hiring decisions involving multiple interdependent candidate features. The performance advantage of CatBoost in particular can be attributed to its use of ordered boosting and advanced handling of categorical features, which minimizes overfitting and maximizes generalization across unseen candidate profiles [14].

This finding reinforces prior research that identifies tree-based ensembles as particularly effective in HR contexts due to their robustness against multicollinearity and missing values [12]. The relatively lower accuracy of models like Logistic Regression (87%) supports the theoretical position that linear models are often too simplistic for high-dimensional hiring problems where relationships between inputs and outcomes are nonlinear and interactive.

B. Model Evaluation and Comparison

Beyond accuracy, the evaluation of precision, recall, and F1-score further strengthens the credibility of CatBoost as the most balanced model. It consistently demonstrated high performance across both classes — hires and non-hires — with minimal skew toward either false positives or false negatives. Random Forest and XGBoost also performed well, though both showed slightly reduced recall for the "hired" class. This implies a greater tendency to overlook suitable candidates, a critical concern in hiring systems where the cost of missing high-potential applicants can be substantial.

The implementation of 30-Fold Cross-Validation added additional statistical rigor by ensuring that every candidate in the dataset contributed to both training and validation phases. The low standard deviation across folds (CatBoost: $\pm 1.2\%$) indicates stable performance, validating the generalizability of the models. This methodological approach aligns with best practices in machine learning research, where cross-validation is essential to reduce variance and overfitting [13].

C. ROC-AUC Curve

While accuracy offers a useful baseline, ROC-AUC serves as a more robust metric, especially for imbalanced datasets. The AUC score of 0.93 for CatBoost, and similarly high scores for Random Forest and XGBoost, indicate excellent class-separating capability. This suggests that the models can effectively differentiate between positive (hired) and negative (not hired) cases across a range of decision thresholds.

Such findings are consistent with prior empirical work that positions AUC as a key indicator of classifier quality in sensitive domains, including finance, medicine, and HR, where the cost of false negatives and false positives is asymmetrical [15]. The fact that all top models achieved AUC values above 0.90 strengthens confidence in their ability to support fair and efficient decision-making in real-world hiring applications.

D. Hyperparameter Tuning

Hyperparameter tuning was conducted to improve model calibration. Interestingly, CatBoost showed remarkable stability, with its accuracy increasing only marginally (from 0.9533 to 0.9542), suggesting that it performs well with default settings and requires minimal parameter adjustment. This property is particularly valuable in enterprise contexts, where computational resources and tuning expertise may be limited.

In contrast, Random Forest and XGBoost experienced slight performance declines post-tuning, likely due to overfitting on specific hyperparameter configurations. Logistic Regression's minimal performance change reflects its simpler structure and fewer tunable parameters. These findings suggest that while tuning remains important, models like CatBoost offer low-risk, high-reward out-of-the-box performance that enhances scalability and adoption.

E. Feature Importance and Interpretability

The use of SHAP values enabled a detailed analysis of feature influence on model predictions. RecruitmentStrategy emerged as the most impactful variable (SHAP: 1.8872), suggesting that organizational policies and hiring intensity play a pivotal role in determining candidate outcomes. This finding aligns with HR theory that institutional structures and hiring frameworks shape recruitment patterns beyond individual qualifications [1].

Other highly influential features included EducationLevel (0.8064), SkillScore (0.8038), and PersonalityScore (0.7793) — indicating that cognitive, academic, and interpersonal traits all contribute meaningfully to hiring decisions. The relatively low SHAP values for Gender (0.1004) and Age (0.0984) are

promising, as they reflect reduced dependency on protected attributes, supporting regulatory alignment and fairness goals.

On the other hand, insights from LIME showed SkillScore (0.1575) as the most locally influential feature, with PreviousCompanies and Gender also contributing to individual predictions. The contrasting rankings between SHAP and LIME underscore the importance of combining both global and local interpretability tools to deliver a more holistic view of model behavior.

F. Fairness & Bias Mitigation

Ensuring algorithmic fairness is essential for the ethical application of machine learning in hiring decisions, where outcomes directly influence people's access to livelihood and opportunity. In this study, fairness was assessed using group fairness metrics—Demographic Parity Difference (DPD) and Equalized Odds Difference (EOD)—which capture disparities in both selection rates and error distribution across gender groups. While the baseline model showed only modest disparities (DPD = 0.0142, EOD = 0.0414), these values are theoretically significant. As noted by Barocas et al. [5], even small gaps, when scaled across thousands of decisions, can reinforce structural inequalities. This reflects the broader concern that machine learning systems trained on historical data may learn and perpetuate past biases, thus unintentionally automating discrimination unless actively corrected.

To mitigate these risks, the study implemented the Exponentiated Gradient algorithm with a demographic parity constraint, which recalibrated the optimization objective to balance both performance and fairness. The result was a measurable reduction in bias metrics (DPD $\sim$ 0.01; EOD $\sim$ 0.03) without a meaningful decline in classification accuracy. This reinforces the theoretical claim that fairness and utility can coexist in algorithmic design when fairness is embedded into the training process [6][9]. Importantly, the success of these fairness-aware adjustments speaks to a growing consensus in the field: that fairness should not be treated as an afterthought or trade-off, but rather as a core design criterion for responsible AI. As such, the fairness evaluation in this study serves not only as an empirical validation step but also as a normative safeguard—ensuring that technological advancement does not entrench social inequality but instead offers a more inclusive and accountable framework for decision-making.

G. Challenges Faced

1) *High Performance Overlap*: A major challenge in this study was the high degree of performance similarity among top models—namely CatBoost, Random Forest, and XGBoost—all achieving accuracies above 93% and ROC-AUC scores exceeding 0.92. While this consistency validates the reliability of ensemble learning for recruitment prediction, it also creates ambiguity in model selection when differences are statistically marginal. As a result, performance metrics alone were insufficient to justify deployment, requiring the incorporation of fairness metrics, interpretability tools, and generalizability tests to make more holistic decisions.

2) *Hyperparameter Tuning*: Hyperparameter optimization, though necessary, presented diminishing returns. For example, tuning improved CatBoost's accuracy only marginally (0.9533 $\rightarrow$ 0.9542), while XGBoost and Random Forest saw slight performance declines, potentially due to overfitting. This suggests that extensive tuning may not always yield proportional gains, especially for already robust models. The computational cost and effort required also highlight a practical tension between theoretical optimization and real-world efficiency.

3) *Contradictions between SHAP and LIME*: An unexpected analytical challenge arose from the divergence between global explanations from SHAP and local explanations from LIME. While SHAP identified RecruitmentStrategy as the most globally influential feature (SHAP impact: 1.8872), LIME ranked it among the least locally impactful (0.0039). Conversely, features like SkillScore were ranked highly by LIME but less dominantly by SHAP. This contradiction illustrates the epistemological difference between the two interpretability approaches: SHAP evaluates feature impact across the entire model space, providing additive, consistent attributions grounded in game theory, whereas LIME approximates feature influence within a localized linear region of a specific instance [17][18]. Interpreting these differences required careful theoretical framing and highlights the importance of using multiple interpretability methods to avoid over-reliance on a single lens and to gain a comprehensive understanding of model behavior in high-stakes domains like recruitment.

H. Practical Implications

1) *Streamlined recruitment workflows*: The automation of early-stage screening using ML can significantly reduce time-to-hire and recruiter workload, allowing HR professionals to reallocate time toward more strategic functions. High-performing models like CatBoost provide accurate and consistent recommendations, helping organizations scale recruitment processes while maintaining quality and consistency.

2) *Bias mitigation in decision making*: Fairness-aware training enables models to make equitable decisions across gender lines, addressing a long-standing issue in algorithmic hiring. By explicitly minimizing disparities

in outcome and error rates, these models align with legal and ethical standards for diversity, equity, and inclusion (DEI). This provides organizations not only with operational benefits, but also reputational and compliance advantages in a regulatory climate increasingly focused on AI ethics.

3) *Explainability and accountability*: Tools like SHAP and LIME empower recruiters and stakeholders to understand and trust model outputs, supporting transparency and auditability. These interpretability frameworks bridge the gap between technical models and human understanding—critical for adoption in real-world HR settings where decisions must often be explained to candidates, auditors, or legal teams.

I. Limitations in this study

1) *Dataset scale and diversity*: The dataset used in this study comprised 1,500 candidate records—adequate for initial model development but limited for capturing the diversity and variance present in real-world recruitment pipelines. The findings, while promising, may not generalize across industries, geographies, or roles without further validation on larger, multi-source datasets.

2) *Feature scope and subjectivity*: The features used primarily reflect structured data—demographics, scores, and qualifications. However, hiring decisions are often influenced by subjective or qualitative factors, such as cultural fit, adaptability, and team dynamics. The exclusion of such features may limit the model’s ability to fully reflect human decision-making processes.

3) *Interpretability limitations in deep learning*: While SHAP and LIME provide transparency for tree-based models, the black-box nature of deep neural networks remains a challenge. Despite achieving strong performance, the DNN used in this study lacked the same level of interpretability, limiting its practical applicability in regulated or high stakes hiring environments.

4) *Historical bias in training data*: Finally, because the models were trained on historical data, there is an inherent risk of perpetuating past biases, even after mitigation efforts. While fairness-aware retraining helped reduce measurable disparities, deeper structural inequalities embedded in the dataset may still influence predictions in subtle ways—an issue requiring ongoing monitoring and ethical oversight.

V. CONCLUSIONS

This study demonstrated the powerful potential of machine learning to optimize hiring decisions while simultaneously addressing concerns of fairness, interpretability, and efficiency. Among the evaluated models, CatBoost emerged as the top performer, achieving a 95.4% accuracy and the highest ROC-AUC score (0.93), while maintaining balanced precision and recall across hiring classes. Random Forest and XGBoost also delivered strong results but exhibited minor asymmetries in recall for the minority class, indicating a tendency to underrepresent successful hiring predictions for less frequent candidate profiles. The incorporation of 30-fold cross-validation further validated the stability and generalizability of these results [1].

A notable contribution of this study lies in its ethical orientation. The use of fairness-aware algorithms—specifically, the Exponentiated Gradient method—successfully mitigated gender-based prediction disparities, reducing Demographic Parity and Equalized Odds differences without sacrificing model accuracy. This supports recent findings that fairness constraints can be integrated into learning objectives without significant loss in performance [2][3]. Additionally, the dual-layered interpretability approach using SHAP and LIME provided valuable global and local insights into model behavior [4][5], affirming that features such as RecruitmentStrategy, EducationLevel, and SkillScore play pivotal roles in hiring outcomes. These contributions offer both theoretical and practical value, proving that machine learning in HR does not have to be a trade-off between performance and fairness. Instead, it can be a strategic enabler of equitable, data-driven hiring systems that align with modern compliance and DEI standards [6].

AREAS FOR FUTURE RESEARCH AND IMPROVEMENTS

Although the current findings are encouraging, there are numerous avenues for future research to further develop and enhance this work.

A. Broader algorithmic exploration

Future studies could explore more complex architectures, such as transformer-based models or explainable deep ensembles, which have shown success in capturing rich, multi-modal patterns in similar domains [7][8].

B. Expanded feature sets

Incorporating unstructured and behavioural data—such as resume text, psychometric test results, social media activity, or video interview analysis—could significantly enhance predictive depth and contextual accuracy [9][19].

C. Deeper fairness modelling

While this study addressed group fairness, future work could explore individual fairness (ensuring similar individuals are treated similarly), intersectional fairness (across multiple sensitive attributes), and longitudinal fairness, tracking long-term impacts of model decisions on organizational diversity and equity [3][10].

D. Greater interpretability for deep learning

Advanced methods such as Integrated Gradients, TCAV, and attention-based visualizations offer promise in interpreting neural networks and should be applied to enhance transparency in regulated HR environments [11].

E. Real-world deployment and feedback loops

Future iterations should include real-time deployment in live hiring systems, with continuous monitoring of downstream outcomes (e.g., time-to-hire, employee performance, retention). This will enable not only performance tracking but also the study of human-AI collaboration dynamics, which are crucial to responsible AI integration [12][13].

F. Cross-context generalization

To improve model robustness and fairness across settings, future research should validate models on larger, multilingual, and culturally diverse datasets, enabling adaptation across industries and geographies [14].

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